Task 1 – Data Modeling

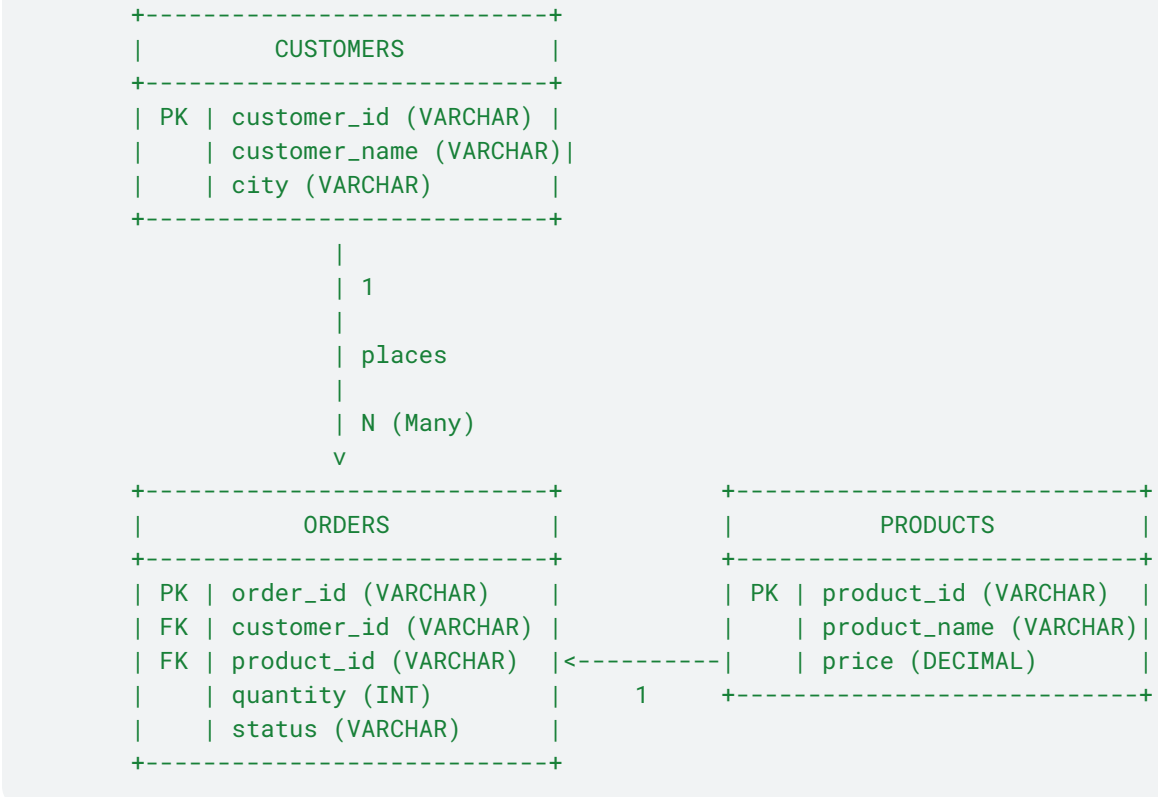
1. Fact Table and Dimension Tables

- **Fact Table:** `Orders` (Contains measurable, quantitative data about business transactions, such as quantity).
- **Dimension Tables:**
  - `Customers` (Contains descriptive attributes about customers: `customer_id`, `customer_name`, `city`).
  - `Products` (Contains descriptive attributes about products: `product_id`, `product_name`, `price`).

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2. Relationship Diagram (Schema Structure)

None



Cardinality & Relationships:

- **Customers (1) to Orders (Many):** One customer can place multiple orders (`1 : N`). Each order belongs to exactly one customer.
- **Products (1) to Orders (Many):** One product can appear across multiple orders (`1 : N`). Each order line item references one product.
- **Architecture Pattern:** A classic **Star Schema** where the central Fact Table (`Orders`) connects to two Dimension Tables (`Customers` and `Products`).

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3. Primary Keys and Foreign Keys

- **`Customers` Table:**
  - **Primary Key:** `customer_id`
- **`Products` Table:**
  - **Primary Key:** `product_id`
- **`Orders` Table:**
  - **Primary Key:** `order_id`
  - **Foreign Keys:** `customer_id` (references `Customers.customer_id`), `product_id` (references `Products.product_id`)

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4. The Grain of the Orders Fact Table

- **Grain Definition:** One row in the `Orders` fact table represents **a single line-item order placed by a customer for a specific product**, including the quantity ordered and the current status of that order.